

```

> restart: unprotect('D'):

m := 9;

twist := - 3;

d := 8;          degree := d + 1:

##

HUU := X^(d+1) + T^(d+1);

#HUU := X^(d+1) + T^(d+1) + 1*X^floor((d+1)/2)*T^(d+1-floor((d+1)
/2));

##

UU := subs({T=1,X=x}, HUU);

##

Jetyx := expand(
+ (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1)^j *
Delta_yx^0, j=0..m-0*3)
+ (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1)^j *
Delta_yx^1, j=0..m-1*3)
+ (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1)^j *
Delta_yx^2, j=0..m-2*3)
+ (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1)^j *
Delta_yx^3, j=0..m-3*3)
) :

##

dA := (m-0*2)*d - m + twist:
dB := (m-1*2)*d - m + twist:
dC := (m-2*2)*d - m + twist:
dD := (m-3*2)*d - m + twist:

##

Delta_yx := y1*x2 - y2*x1:

Soly1 := - x1*Rx/Ry - z1*Rz/Ry:

```

```
Soly2 := -x1^2*Rxx/Ry+2*x1^2*Rxy*Rx/Ry^2+2*x1*Rxy*z1*Rz/Ry^2-2*
x1*z1*Rxz/Ry-Ryy*x1^2*Rx^2/Ry^3-2*Ryy*x1*Rx*z1*Rz/Ry^3-Ryy*z1^2*
Rz^2/Ry^3+2*z1*Ryz*x1*Rx/Ry^2+2*z1^2*Ryz*Rz/Ry^2-z1^2*Rzz/Ry-x2*
Rx/Ry-z2*Rz/Ry:
```

```
##
```

```
Jetzx := sort(mtaylor(
                                expand(subs(x2=(Deltazx+x1*z2)/z1,
                                expand(subs({y1=Soly1,y2=Soly2},
Jetyx))))),
                                [x1,z1,Deltazx], 100), [x1,z1,Deltazx]):
```

```
##
```

```
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
EEEq[m-j-3*k,j,k] := factor(
    Rz^(m-j-3*k)
*
    Ry^(m-2*k)
*
    coeftayl(Jetzx, [x1,z1,Deltazx]=[0,0,0], [m-j-3*k,j,k])):
od: od:
```

```
## EQUATION RR EXPLICITE
```

```
RR := z^(d+1) + y^(d+1) + UU;
```

```
Rx := diff(RR,x):
Ry := diff(RR,y):
Rz := diff(RR,z):
```

```
Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Rxz := diff(RR,x,z):
Ryy := diff(RR,y,y):
Ryz := diff(RR,y,z):
Rzz := diff(RR,z,z):
```

```
##
```

```
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
od: od:
```

```
##
```

```
for j1 from 0 to 1+m*d+d do
```

```
ny[j1] := (-z^(d+1)-UU)^(floor(j1/(d+1))) * y^(j1-(d+1)*floor(j1/
(d+1))):
```

```
od:
```

$$m := 9$$

$$twist := -3$$

$$d := 8$$

$$HUU := T^9 + X^9$$

$$UU := x^9 + 1$$

$$RR := x^9 + y^9 + z^9 + 1 \quad (1)$$

```
> ##
```

```
for l from 0 to m-0*3 do AA||l := add(A||l[j]*y^j, j=0..min(d,
dA)): od:
```

```
for l from 0 to m-1*3 do BB||l := add(B||l[j]*y^j, j=0..min(d,
dB)): od:
```

```
for l from 0 to m-2*3 do CC||l := add(C||l[j]*y^j, j=0..min(d,
dC)): od:
```

```
for l from 0 to m-3*3 do DD||l := add(D||l[j]*y^j, j=0..min(d,
dD)): od:
```

```
##
```

```
1 + m*d + d:
```

```
printlevel := 2:
```

```
for k from 0 to m/3 do
```

```
for j from 0 to m-3*k do
```

```
degree(EEEq[m-j-3*k,j,k], y):
```

```
EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], y, 1+m*d+d):
```

```
dy[m-j-3*k,j,k] := degree(EEEq[m-j-3*k,j,k], y):
```

```
od: od:
```

```
##
```

```
printlevel := 3:
```

```
for k from 0 to m/3 do
```

```

for j from 0 to m-3*k do
for j1 from 0 to dy[m-j-3*k,j,k] do
Cy[m-j-3*k,j,k,j1] := coeff(EEEq[m-j-3*k,j,k], y, j1):
od:
od: od:
##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := add(Cy[m-j-3*k,j,k,j1]*ny[j1], j1=0..dy[m-
j-3*k,j,k]):

degree(EEq[m-j-3*k,j,k], y):
od: od:
##

printlevel := 2:
for l from 0 to m-0*3 do
for j from 0 to min(d,dA) do

A||l[j] := add(A||l[j,k]*z^k, k=0..dA-j):

od: od:

printlevel := 2:
for l from 0 to m-1*3 do
for j from 0 to min(d,dB) do

B||l[j] := add(B||l[j,k]*z^k, k=0..dB-j):

od: od:

printlevel := 2:
for l from 0 to m-2*3 do
for j from 0 to min(d,dC) do

C||l[j] := add(C||l[j,k]*z^k, k=0..dC-j):

od: od:

printlevel := 2:
for l from 0 to m-3*3 do
for j from 0 to min(d,dD) do

D||l[j] := add(D||l[j,k]*z^k, k=0..dD-j):

```

```

od: od:

##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
degree(EEq[m-j-3*k,j,k], z):

od: od:

##

printlevel := 3:
for l from 0 to m-0*3 do
for j from 0 to min(d,dA) do
for k from 0 to dA-j do

A||l[j,k] := add(A||l[i,j,k]*x^i, i=0..dA-j-k):

od: od: od:

printlevel := 3:
for l from 0 to m-1*3 do
for j from 0 to min(d,dB) do
for k from 0 to dB-j do

B||l[j,k] := add(B||l[i,j,k]*x^i, i=0..dB-j-k):

od: od: od:

printlevel := 3:
for l from 0 to m-2*3 do
for j from 0 to min(d,dC) do
for k from 0 to dC-j do

C||l[j,k] := add(C||l[i,j,k]*x^i, i=0..dC-j-k):

od: od: od:

printlevel := 3:
for l from 0 to m-3*3 do
for j from 0 to min(d,dD) do
for k from 0 to dD-j do

D||l[j,k] := add(D||l[i,j,k]*x^i, i=0..dD-j-k):

od: od: od:

##

```

```

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

```
##
```

```

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

degree(Eq[m-j-3*k,j,k], z):

```

```
od: od:
```

```
##
```

```

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

degree(Eq[m-j-3*k,j,k], y):

```

```
od: od;
```

```
l := 0
```

```
14907
```

```
l := 1
```

```
14907
```

```
l := 2
```

```
14907
```

```
l := 3
```

```
14907
```

```
l := 4
```

```
14907
```

```
l := 5
```

```
14907
```

```
l := 6
```

```
14907
```

```
l := 7
```

```
14907
```

```
l := 8
```

```
14907
```

```
l := 9
```

14907

$l := 10$

$l := 0$

7779

$l := 1$

7779

$l := 2$

7779

$l := 3$

7779

$l := 4$

7779

$l := 5$

7779

$l := 6$

7779

$l := 7$

$l := 0$

2955

$l := 1$

2955

$l := 2$

2955

$l := 3$

2955

$l := 4$

$l := 0$

435

$l := 1$

$k := 0$

8

8

8

8

8

8

8

8

$$\begin{aligned} & 8 \\ & - \infty \\ & k := 1 \\ & 8 \\ & 8 \\ & 8 \\ & 8 \\ & 8 \\ & 8 \\ & - \infty \\ & k := 2 \\ & 8 \\ & 8 \\ & 8 \\ & - \infty \\ & k := 3 \\ & - \infty \\ & k := 4 \end{aligned}$$

(2)

```
> ## SYSTEM FOR DIVISIBILITY BY z^(1*d)
Systemez1d := {}:
for k from 0 to (m-1)/3 do
  Coeffs_z1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], z, 1*d))), [x,y,z]}:
  Systemez1d := Systemez1d union Coeffs_z1d[1,m-1-3*k,k]:
od:
Systeme_z1d := Systemez1d:
## VARIABLES FOR DIVISIBILITY BY z^(1*d)
```



```

Variables_z1d := indets(Systeme_z1d):
## RESOLUTION z^(1*d)
Solutions_z1d := solve(Systeme_z1d, Variables_z1d):
assign(Solutions_z1d):
##
for l from 0 to m - 0*3 do nops(expand(AA||1)) od;
for l from 0 to m - 1*3 do nops(expand(BB||1)) od;
for l from 0 to m - 2*3 do nops(expand(CC||1)) od;
for l from 0 to m - 3*3 do nops(expand(DD||1)) od;

```

l := 0

14907

l := 1

14907

l := 2

14907

l := 3

14907

l := 4

14907

l := 5

14907

l := 6

14907

l := 7

14907

l := 8

14439

l := 9

18183

l := 10

l := 0

7779

l := 1

7779

l := 2

7779

```

l := 3
7779
l := 4
7779
l := 5
7302
l := 6
9512
l := 7
l := 0
2955
l := 1
2955
l := 2
2459
l := 3
3341
l := 4
l := 0
435
l := 1

```

(3)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(2*d)}$ 
Systemez2d := {}:
for k from 0 to (m-2)/3 do
  Coeffs_z2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], z, 2*d)), [x,y,z])}:
Systemez2d := Systemez2d union Coeffs_z2d[2,m-2-3*k,k]:
od:
Systeme_z2d := Systemez2d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(2*d)}$ 
Variables_z2d := indets(Systeme_z2d):
## RESOLUTION  $z^{(2*d)}$ 

```

```
Solutions_z2d := solve(Systeme_z2d, Variables_z2d):
```

```
assign(Solutions_z2d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

```
l := 0
```

```
14907
```

```
l := 1
```

```
14907
```

```
l := 2
```

```
14907
```

```
l := 3
```

```
14907
```

```
l := 4
```

```
14907
```

```
l := 5
```

```
14907
```

```
l := 6
```

```
14907
```

```
l := 7
```

```
14471
```

```
l := 8
```

```
17323
```

```
l := 9
```

```
31079
```

```
l := 10
```

```
l := 0
```

```
7779
```

```
l := 1
```

```
7779
```

```
l := 2
```

```
7779
```

```
l := 3
```

```
7779
```

```
l := 4
```

7103

$l := 5$

8637

$l := 6$

15370

$l := 7$

$l := 0$

2955

$l := 1$

1342

$l := 2$

2254

$l := 3$

2867

$l := 4$

$l := 0$

525

$l := 1$

(4)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(3*d)}$ 
Systemez3d := {}:
for k from 0 to (m-3)/3 do
  Coeffs_z3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], z, 3*d)), [x,y,z])}:
Systemez3d := Systemez3d union Coeffs_z3d[3,m-3-3*k,k]:
od:
Systeme_z3d := Systemez3d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(3*d)}$ 
Variables_z3d := indets(Systeme_z3d):
## RESOLUTION  $z^{(3*d)}$ 
Solutions_z3d := solve(Systeme_z3d, Variables_z3d):
assign(Solutions_z3d):
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

l := 0

14907

l := 1

14907

l := 2

14907

l := 3

14907

l := 4

14907

l := 5

14907

l := 6

16675

l := 7

19292

l := 8

28621

l := 9

54853

l := 10

l := 0

7779

l := 1

7779

l := 2

7779

l := 3

6096

l := 4

7772

l := 5

11117

l := 6

19131

$l := 7$

$l := 0$

331

$l := 1$

807

$l := 2$

937

$l := 3$

1062

$l := 4$

$l := 0$

502

$l := 1$

(5)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(4*d)}$ 
Systemez4d := {}:
for k from 0 to (m-4)/3 do
  Coeffs_z4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], z, 4*d)), [x,y,z])}:
Systemez4d := Systemez4d union Coeffs_z4d[4,m-4-3*k,k]:
od:
Systeme_z4d := Systemez4d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(4*d)}$ 
Variables_z4d := indets(Systeme_z4d):
## RESOLUTION  $z^{(4*d)}$ 
Solutions_z4d := solve(Systeme_z4d, Variables_z4d):
assign(Solutions_z4d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$l := 0$
14907
 $l := 1$
14907
 $l := 2$
14907
 $l := 3$
14907
 $l := 4$
14907
 $l := 5$
20809
 $l := 6$
20957
 $l := 7$
29379
 $l := 8$
42413
 $l := 9$
81474
 $l := 10$
 $l := 0$
7779
 $l := 1$
7779
 $l := 2$
3887
 $l := 3$
7100
 $l := 4$
10236
 $l := 5$
12733
 $l := 6$
16817
 $l := 7$
 $l := 0$
529

```

l := 1
1345
l := 2
1542
l := 3
1642
l := 4
l := 0
872
l := 1

```

(6)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(5*d)}$ 
Systemez5d := {}:
for k from 0 to (m-5)/3 do
  Coeffs_z5d[5,m-5-3*k,k] := {coeffs(expand(mTaylor(Eq[5,m-5-3*k,
k], z, 5*d)), [x,y,z])}:
Systemez5d := Systemez5d union Coeffs_z5d[5,m-5-3*k,k]:
od:
Systeme_z5d := Systemez5d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(5*d)}$ 
Variables_z5d := indets(Systeme_z5d):
## RESOLUTION  $z^{(5*d)}$ 
Solutions_z5d := solve(Systeme_z5d, Variables_z5d):
assign(Solutions_z5d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

l := 0
14907
l := 1
14907

```


$l := 2$
14907
 $l := 3$
14907
 $l := 4$
14534
 $l := 5$
23970
 $l := 6$
24820
 $l := 7$
36734
 $l := 8$
54423
 $l := 9$
79390
 $l := 10$
 $l := 0$
7779
 $l := 1$
1003
 $l := 2$
2241
 $l := 3$
3119
 $l := 4$
3958
 $l := 5$
4499
 $l := 6$
4054
 $l := 7$
 $l := 0$
636
 $l := 1$
1138
 $l := 2$
1412

$l := 3$

1779

$l := 4$

$l := 0$

512

$l := 1$

(7)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(6*d)}$ 
Systemez6d := {}:
for k from 0 to (m-6)/3 do
  Coeffs_z6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
k], z, 6*d)), [x,y,z])}:
Systemez6d := Systemez6d union Coeffs_z6d[6,m-6-3*k,k]:
od:
Systeme_z6d := Systemez6d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(6*d)}$ 
Variables_z6d := indets(Systeme_z6d):
## RESOLUTION  $z^{(6*d)}$ 
Solutions_z6d := solve(Systeme_z6d, Variables_z6d):
assign(Solutions_z6d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

 $l := 0$ 
14907
 $l := 1$ 
14907
 $l := 2$ 
14907
 $l := 3$ 
6134
```

$l := 4$
11378
 $l := 5$
18581
 $l := 6$
20792
 $l := 7$
27329
 $l := 8$
35151
 $l := 9$
46167
 $l := 10$
 $l := 0$
1
 $l := 1$
71
 $l := 2$
264
 $l := 3$
326
 $l := 4$
310
 $l := 5$
127
 $l := 6$
1
 $l := 7$
 $l := 0$
71
 $l := 1$
266
 $l := 2$
302
 $l := 3$
165
 $l := 4$
 $l := 0$

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(7*d)}$ 

Systemez7d := {}:

for k from 0 to (m-7)/3 do

  Coeffs_z7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], z, 7*d)), [x,y,z])}:

  Systemez7d := Systemez7d union Coeffs_z7d[7,m-7-3*k,k]:

od:

Systeme_z7d := Systemez7d:

## VARIABLES FOR DIVISIBILITY BY  $z^{(7*d)}$ 

Variables_z7d := indets(Systeme_z7d):

## RESOLUTION  $z^{(7*d)}$ 

Solutions_z7d := solve(Systeme_z7d, Variables_z7d):

assign(Solutions_z7d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

l := 0
14907
l := 1
14907
l := 2
600
l := 3
1375
l := 4
2006
l := 5
3117

```

$l := 6$

4025

$l := 7$

4784

$l := 8$

5307

$l := 9$

5541

$l := 10$

$l := 0$

1

$l := 1$

56

$l := 2$

150

$l := 3$

188

$l := 4$

199

$l := 5$

127

$l := 6$

1

$l := 7$

$l := 0$

56

$l := 1$

170

$l := 2$

187

$l := 3$

144

$l := 4$

$l := 0$

63

$l := 1$

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(8*d)}$ 

Systemez8d := {}:
for k from 0 to (m-8)/3 do
  Coeffs_z8d[8,m-8-3*k,k] := {coeffs(expand(mTaylor(Eq[8,m-8-3*k,
k], z, 8*d)), [x,y,z])}:
Systemez8d := Systemez8d union Coeffs_z8d[8,m-8-3*k,k]:
od:
Systeme_z8d := Systemez8d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(8*d)}$ 
Variables_z8d := indets(Systeme_z8d):
## RESOLUTION  $z^{(8*d)}$ 
Solutions_z8d := solve(Systeme_z8d, Variables_z8d):
assign(Solutions_z8d):
##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

l := 0
14907
l := 1
1
l := 2
1
l := 3
20
l := 4
20
l := 5
20
l := 6
20
l := 7
1

```

```

l := 8
1
l := 9
1
l := 10
l := 0
1
l := 1
1
l := 2
20
l := 3
20
l := 4
20
l := 5
1
l := 6
1
l := 7
l := 0
1
l := 1
20
l := 2
20
l := 3
1
l := 4
l := 0
20
l := 1

```

(10)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(9*d)}$ 
Systemez9d := {}:
for k from 0 to (m-9)/3 do

```

```
Coeffs_z9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], z, 9*d)), [x,y,z])}:

```

```
Systemez9d := Systemez9d union Coeffs_z9d[9,m-9-3*k,k]:

```

```
od:

```

```
Systeme_z9d := Systemez9d:

```

```
## VARIABLES FOR DIVISIBILITY BY  $z^{(9*d)}$ 

```

```
Variables_z9d := indets(Systeme_z9d):

```

```
## RESOLUTION  $z^{(9*d)}$ 

```

```
Solutions_z9d := solve(Systeme_z9d, Variables_z9d):

```

```
assign(Solutions_z9d):

```

```
##

```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

$l := 0$

1

$l := 1$

1

$l := 2$

1

$l := 3$

20

$l := 4$

20

$l := 5$

20

$l := 6$

20

$l := 7$

1

$l := 8$

1

$l := 9$

1

$l := 10$

$l := 0$

1

$l := 1$

1

$l := 2$

20

$l := 3$

20

$l := 4$

20

$l := 5$

1

$l := 6$

1

$l := 7$

$l := 0$

1

$l := 1$

20

$l := 2$

20

$l := 3$

1

$l := 4$

$l := 0$

20

$l := 1$

(11)

> ## SYSTEM FOR DIVISIBILITY BY $z^{(10*d)}$

Systemez10d := {}:

for k from 0 to $(m-10)/3$ do

Coeffs_z10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10-3*k,k], z, $10*d$)), [x,y,z])}:}

Systemez10d := Systemez10d union Coeffs_z10d[10,m-10-3*k,k]:

```

od:
Systeme_z10d := Systemez10d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(10*d)}$ 
Variables_z10d := indets(Systeme_z10d):
## RESOLUTION  $z^{(10*d)}$ 
Solutions_z10d := solve(Systeme_z10d, Variables_z10d):
assign(Solutions_z10d):
##
for l from 0 to m - 0*3 do nops(expand(AA||1)) od;
for l from 0 to m - 1*3 do nops(expand(BB||1)) od;
for l from 0 to m - 2*3 do nops(expand(CC||1)) od;
for l from 0 to m - 3*3 do nops(expand(DD||1)) od;

l := 0
1
l := 1
1
l := 2
1
l := 3
20
l := 4
20
l := 5
20
l := 6
20
l := 7
1
l := 8
1
l := 9
1
l := 10
l := 0
1
l := 1

```

```

1
l := 2
20
l := 3
20
l := 4
20
l := 5
1
l := 6
1
l := 7
l := 0
1
l := 1
20
l := 2
20
l := 3
1
l := 4
l := 0
20
l := 1

```

(12)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(11*d)}$ 
Systemez11d := {}:
for k from 0 to (m-11)/3 do
  Coeffs_z11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11-3*k,k], z, 11*d)), [x,y,z])}:
Systemez11d := Systemez11d union Coeffs_z11d[11,m-11-3*k,k]:
od:
Systeme_z11d := Systemez11d:
## VARIABLES FOR DIVISIBILITY BY  $z^{(11*d)}$ 

```

```

Variables_z11d := indets(Systeme_z11d):
## RESOLUTION  $z^{(11*d)}$ 
Solutions_z11d := solve(Systeme_z11d, Variables_z11d):
assign(Solutions_z11d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

$l := 0$

1

$l := 1$

1

$l := 2$

1

$l := 3$

20

$l := 4$

20

$l := 5$

20

$l := 6$

20

$l := 7$

1

$l := 8$

1

$l := 9$

1

$l := 10$

$l := 0$

1

$l := 1$

1

$l := 2$

20

$l := 3$

```

20
l := 4
20
l := 5
1
l := 6
1
l := 7
l := 0
1
l := 1
20
l := 2
20
l := 3
1
l := 4
l := 0
20
l := 1

```

(13)

```

> ## HOLOMORPHICITY AT INFINITY

for l from 0 to m - 0*3 do

HA||l := add(add(add(A||l[i,j,k]*X^i*Z^k*Y^j*T^(dA-i-j-k), i=0..
dA-j-k), k=0..dA-j), j=0..min(d,dA)):

od:

#

for l from 0 to m - 1*3 do

```

```

HB||l := add(add(add(B||l[i,j,k]*X^i*Z^k*Y^j*T^(dB-i-j-k), i=0..
dB-j-k), k=0..dB-j), j=0..min(d,dB)):

od:

#

for l from 0 to m - 2*3 do

HC||l := add(add(add(C||l[i,j,k]*X^i*Z^k*Y^j*T^(dC-i-j-k), i=0..
dC-j-k), k=0..dC-j), j=0..min(d,dC)):

od:

#

for l from 0 to m - 3*3 do

HD||l := add(add(add(D||l[i,j,k]*X^i*Z^k*Y^j*T^(dD-i-j-k), i=0..
dD-j-k), k=0..dD-j), j=0..min(d,dD)):

od:

##

for h from 0 to m - 3*0 do

HAEEq[h] := add(HA||l*binomial(l,h)*X^(m-3*0-l)*Y^(l-h), l=h..
m-3*0):
dYA[h] := degree(HAEEq[h], Y):

od:

#

for h from 0 to m - 3*1 do

HBEEq[h] := add(HB||l*binomial(l,h)*X^(m-3*1-l)*Y^(l-h), l=h..
m-3*1):
dYB[h] := degree(HBEEq[h], Y):

od:

#

for h from 0 to m - 3*2 do

HCEEq[h] := add(HC||l*binomial(l,h)*X^(m-3*2-l)*Y^(l-h), l=h..
m-3*2):
dYC[h] := degree(HCEEq[h], Y):

od:

#

for h from 0 to m - 3*3 do

```

```

HDEEq[h] := add(HD||l*binomial(l,h)*X^(m-3*3-l)*Y^(l-h), l=h..
m-3*3):
dYD[h] := degree(HDEEq[h], Y):

```

```

od:

```

```

##

```

```

for h from 0 to m - 3*0 do
for j1 from 0 to dYA[h] do

```

```

CoeffA[h,j1] := coeff(HAEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*1 do
for j1 from 0 to dYB[h] do

```

```

CoeffB[h,j1] := coeff(HBEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*2 do
for j1 from 0 to dYC[h] do

```

```

CoeffC[h,j1] := coeff(HCEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*3 do
for j1 from 0 to dYD[h] do

```

```

CoeffD[h,j1] := coeff(HDEEq[h], Y, j1):

```

```

od: od:

```

```

##

```

```

for j1 from 0 to 1+m*d+d do

```

```

nY[j1] := (-Z^(d+1)-HUU)^(floor(j1/(d+1))) * Y^(j1-(d+1)*floor
(j1/(d+1))):

```

```

od:

```

```

##

```

```

for h from 0 to m - 3*0 do

```

```

HAEq[h] := expand(mtaylor(add(CoeffA[h,j1]*nY[j1], j1=0..dYA[h]),
T, m-3*0-h)):

```

```

degree(HAEq[h],Y):
od:
#
for h from 0 to m - 3*1 do
HBEq[h] := expand(mtaylor(add(CoeffB[h,j1]*nY[j1], j1=0..dYB[h]),
T, m-3*1-h)):
degree(HBEq[h],Y):
od:
#
for h from 0 to m - 3*2 do
HCEq[h] := expand(mtaylor(add(CoeffC[h,j1]*nY[j1], j1=0..dYC[h]),
T, m-3*2-h)):
degree(HCEq[h],Y):
od:
#
for h from 0 to m - 3*3 do
HDEq[h] := expand(mtaylor(add(CoeffD[h,j1]*nY[j1], j1=0..dYD[h]),
T, m-3*3-h)):
degree(HDEq[h],Y):
od:

```

```

> HSysteme := {seq(coeffs(HAEq[h],[T,X,Y,Z]),h=0..m-3*0)}
               union {seq(coeffs(HBEq[h],[T,X,Y,Z]),h=0..m-3*1)}
               union {seq(coeffs(HCEq[h],[T,X,Y,Z]),h=0..m-3*2)}
               union {seq(coeffs(HDEq[h],[T,X,Y,Z]),h=0..m-3*3)}
minus {0}:

HSolutions := solve(HSysteme):

assign(HSolutions):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;

```



```
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

l := 0

1

l := 1

1

l := 2

1

l := 3

4

l := 4

4

l := 5

5

l := 6

4

l := 7

1

l := 8

1

l := 9

1

l := 10

l := 0

1

l := 1

1

l := 2

5

l := 3

5

l := 4

5

l := 5

1

l := 6

1

[

$l := 7$
 $l := 0$
 $\mathbf{1}$
 $l := 1$
 5
 $l := 2$
 5
 $l := 3$
 $\mathbf{1}$
 $l := 4$
 $l := 0$
 5
 $l := 1$

(14)

[> Jetyx := factor(

```

+ (1/Rz^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) * (y1)^j
* Delta_yx^0, j=0..m-0*3)
+ (1/Rz^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) * (y1)^j
* Delta_yx^1, j=0..m-1*3)
+ (1/Rz^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) * (y1)^j
* Delta_yx^2, j=0..m-2*3)
+ (1/Rz^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) * (y1)^j
* Delta_yx^3, j=0..m-3*3)
) :

```

```

> Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
NbVars := nops(Vars);

NbVars := 1
(15)

```

```

> Ens_Vars := {}:
for i from 1 to nops(Vars) do
  Ens_Vars := Ens_Vars union {op(i,Vars)=0}:
od:
EnsVars := Ens_Vars:

```

```

> for i from 1 to nops(Vars) do
  JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jetyx))):
od:

```

```

> ## VERIFICATION OF HOLOMORPHICITY AT INFINITY
Liste_x := {x = 1/x,
            y = y/x,
            z = z/x,
            x1 = - x1/x^2,
            x2 = 2*x1^2/x^3 - x2/x^2,
            y1 = y1/x - y*x1/x^2,
            y2 = y2/x - 2*y1*x1/x^2 + 2*y*x1^2/x^3 - y*x2/x^2}:

##
for i from 1 to nops(Vars) do
JetSol_x[i] := factor(subs(Liste_x, JetSol[i])):
od:

```

```

> Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
NbVars := nops(Vars);

```

NbVars := 1 (16)

```

> ## SEE AGAIN THE SOLUTIONS
for i from 1 to nops(Vars) do
JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jetyx))):
od;

```

i := 1

$$JetSol_1 := \frac{x^3 (xx1yy2 + 8xx1yl^2 - xx2yy1 - 8yl yxl^2)^3}{595077871104 z^{21}}$$

i := 2 (17)

